

Session Objectives

1. Explain the mechanism(s) of action for ADHD medications.
2. Describe the clinical benefits of ADHD drug therapy.
3. Recognize key differences between medications used to treat ADHD.
4. Identify adverse effects related to ADHD pharmacotherapy.
5. Recommend appropriate monitoring parameters for patients on ADHD medications.

ADHD Management

**NEW YORK INSTITUTE
OF TECHNOLOGY**

College of Osteopathic
Medicine

- Pharmacotherapy is only one modality
- Psychosocial treatments
 - Behavioral therapy
 - Training interventions
 - Mindfulness, cognitive training, EEG biofeedback, counseling, dietary modification

- Key considerations regarding pharmacologic treatment:
 - The goal is to control symptoms and behavior that impair learning and development.
 - Individualized and specific behavioral, academic, and social target goals and treatments should be identified.
 - Response to medications is not a paradoxical effect.

- TJ is an 8-year-old boy who is in the 3rd grade. He was recently diagnosed with ADHD after his teacher noticed that he does not follow through on tasks, is easily distracted, is very forgetful, and does not seem to listen.
- Social and behavioral interventions have been initiated at home and school. However, the decision has been made to begin pharmacotherapy since TJ continues to struggle in the classroom.
- What is a reasonable therapy to recommend?

- Stimulants are first-line therapy, unless the patient has certain co-morbidities.
- The stimulant group includes amphetamines & methylphenidate.
 - Approximately 40% respond to both.
 - 40% respond to only one.
 - ADHD subtype does not predict response to specific agent.

- Drug therapy should be chosen based on dosing, duration, side effects, patient's ability to swallow a pill, cost.
- Height, weight, eating and sleeping patterns should be recorded at baseline and every 3 months.
- Initial dose should be titrated to individualize dose and to minimize adverse effects.
 - Can be titrated on a 3-to-7-day basis.
 - Dose may be titrated with a weekly or biweekly telephone call to the family during the first month.
- A face-to face follow-up visit is recommended by the fourth week of medication.
 - Response to medication can be assessed, as well as adverse effects, pulse, blood pressure, and weight.

Stimulants: Adverse Effects

- Typically dose-related
- Insomnia
 - Dose stimulants earlier in the day
 - Less insomnia with IR formulations
- Growth suppression
 - Weight loss – minimal for most children
 - Methylphenidate: ↓ 3 kg in first year, 1.2 kg in 2nd year
 - Primarily due to reduced food intake vs ↑ in metabolism
 - Administer high calorie meal at breakfast or dinner; snacks
 - Products vary regarding administration with food (methylphenidate IR products, take 30-45 minutes before a meal)
- Headache - treat
- Abdominal pain
 - Administer with food; make sure meals are eaten

- Baseline screening recommendations
 - Medical and family history
 - Sudden death, CV symptoms, Wolff-Parkinson-White syndrome, hypertrophic cardiomyopathy, long QT syndrome
 - Avoid stimulants if known structural cardiac abnormalities
 - Baseline electrocardiogram if risk factors present

- Zhang L, Honghui Y, Lin L, et al. Risk of cardiovascular diseases associated with medications used in attention-deficit/hyperactivity disorder: A systematic review and meta-analysis. JAMA Network 2022;5(11):e2243597.
- 19 observational studies investigating associations between ADHD medication use and risk of any CVD across age groups
- Outcomes:
 - No significant association with risk of CVD events across age groups
 - No significant association of CVD risk in subgroup analyses
 - 2/8 studies with long-term follow-up showed increased risk in those with history of CVD

- Tics or abnormal movements
 - A tic is a brief, rapid, recurrent, and seemingly purposeless stereotyped motor contraction (blinking, twitching of the nose, jerking of the neck).
- Priapism
 - Definition: prolonged (> 4 hours) and sometimes painful erection
 - Post-marketing cases with methylphenidate prompted warning in December 2013
 - Mean age of patients has been 12.5 years
 - Has occurred on and off the medication

- Improvement in scholastic performance has been documented in elementary students.
 - Increased reading and math score compared to non-treated patients
 - Scores still lag behind non-ADHD peers
- Scales used at baseline can be evaluation tools for efficacy.
- Conners Comprehensive Behavior Rating Scales (Conners CBRS)
 - One for teacher, parent, child

Stimulants: Lack of Desired Response

- Increase the dose:
 - Off-label high doses – potential increase in cardiac events
- Variation in treatment response
 - Pharmacology of medications
 - Pharmacokinetic variations
 - Pharmacogenetic differences
- Change to a different formulation or to a different class of stimulant
 - Patient history is important – (ex, medication effective but wearing off)
- Dose conversion information is available for switching between products

WARNING: ABUSE, MISUSE, AND ADDICTION

Adderall® has a high potential for abuse and misuse, which can lead to the development of a substance use disorder, including addiction. Misuse and abuse of CNS stimulants, including Adderall®, can result in overdose and death, and this risk is increased with higher doses or unapproved methods of administration, such as snorting or injection.

Before prescribing Adderall®, assess each patient's risk for abuse, misuse, and addiction. Educate patients and their families about these risks, proper storage of the drug, and proper disposal of any unused drug. Throughout Adderall® treatment, reassess each patient's risk of abuse, misuse, and addiction and frequently monitor for signs and symptoms of abuse, misuse, and addiction.

- Alternatives to stimulants if co-morbidities present
- Not as effective as stimulant medications
- Onset is 2-4 weeks vs 1-2 hours with stimulants
- Patients with cardiovascular risk factors at baseline may need further evaluation (electrocardiogram, consult with a cardiologist)

Nonstimulants: Atomoxetine

- Adverse effects:
 - Abdominal pain
 - Somnolence
 - Decreased appetite
 - Cardiovascular effects: ↑ in heart rate and blood pressure
- Metabolized by CYP2D6
 - Poor metabolizers have AUCs that are 10x ↑ and peak concentrations that are 4x ↑ extensive metabolizers
- Rare liver injury may occur
 - Routine monitoring of hepatic labs is not recommended
 - Should be discontinued if signs of liver injury present (pruritus, dark urine, jaundice, right upper quadrant tenderness)
- Warning about suicidal ideation due to mechanism that is similar to antidepressants

Nonstimulants: Alpha 2a Adrenergic Agonists

- MOA: Traditionally defined as *pre-synaptic* alpha-2 agonists (eg, antihypertensive agents); data indicates a *post-synaptic* alpha-2A agonist effects in ADHD
- Post-synaptic effect associated with increased blood flow in prefrontal cortex, which is thought to enhance memory and improve executive functioning.
- Indicated as monotherapy or in combination with stimulants
- Are options for patients who can not tolerate stimulant side effects, or who have co-morbid tics or aggression, high risk for drug dependence
- Guanfacine, extended-release (Intuniv®); can give once daily
- Clonidine, extended-release (Kapvay®)

Nonstimulants: Alpha 2a Adrenergic Agonists

- Adverse effects
- Sedation
 - May be a benefit
- Bradycardia and hypotension
- Potential for rebound hypertension when stopped
- Constipation

FDA Warning: Stimulants, Atomoxetine, Alpha-2 Agonists

- Stimulants, atomoxetine, alpha-2 agonists all have a warning about the potential for certain psychiatric adverse effects.
- Psychiatric cases have been reported with all three classes, but more frequently with the stimulants.
- Three categories of effects:
 - Psychosis – hallucinations involving visual or tactile sensations; delusions
 - Mood disturbances – irritability, depression, labile mood
 - Severe anxiety/panic attacks

- ‘Holidays’ are periods of time off medication
 - Weekends, summertime
- Potential benefits
 - ‘Catch-up’ in growth
 - Re-assessment for continued need of drug therapy
- Benefit must be balanced against any negative impact of stopping medication, such as learning, self-image, socialization
- Atomoxetine and alpha-2 agonists have a more gradual onset of action and longer duration of effect
 - Drug holidays not recommended

Nonstimulants: Bupropion

- Bupropion (Wellbutrin®, Zyban®)
 - MOA: A weak Nor and DA reuptake inhibitor
 - Approved for depression, smoking cessation
 - Used off-label in children, adolescents, and adults
 - Considered an alternative to other agents; benefit inconsistent
 - Adverse effects: appetite suppression (less than stimulants), insomnia, tachycardia; may lower seizure threshold in some patients
 - Class suicidal ideation warning

- Stimulants are Schedule II drugs per the federal Drug Enforcement Agency (DEA)
 - Prescriptions can not be refilled.
 - Criteria for prescribing stimulants in the treatment of ADHD may vary by state.
- When to stop ADHD medications
 - Individualized discussion with patient and/or parents

- Pharmacotherapy is only one management modality.
- Therapeutic goals should be identified before medication is initiated.
- Stimulants are first-line therapy pharmacotherapy for most patients.
- Stimulant doses should be titrated to maximize efficacy and minimize adverse effects.
- Patients should be periodically assessed for efficacy and toxicity of ADHD medications.